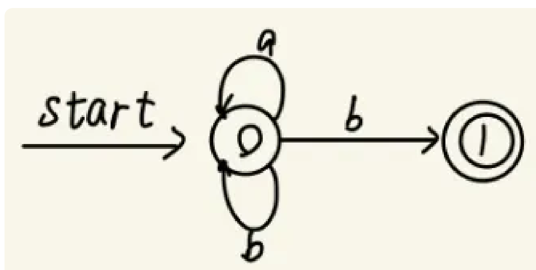
assignment2

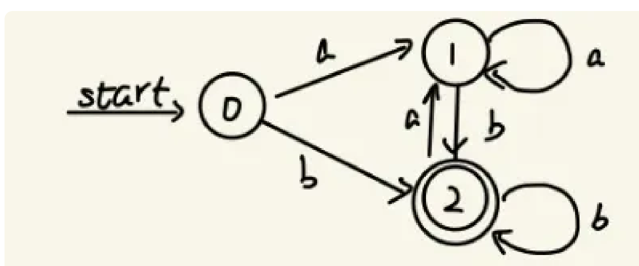
Exercise1

我的方法：可以根据消结律来快速写出NFA，然后用子集构造算法转换成DFA。这可能和汤普森构造算法求得的NFA不同，但是等价NFA对应同一个DFA。

1. L((a|b)*b)：由字符{a,b}构成且以b结尾的字符串。

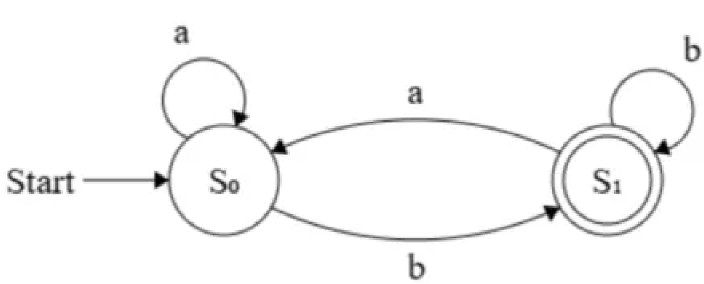


NFA

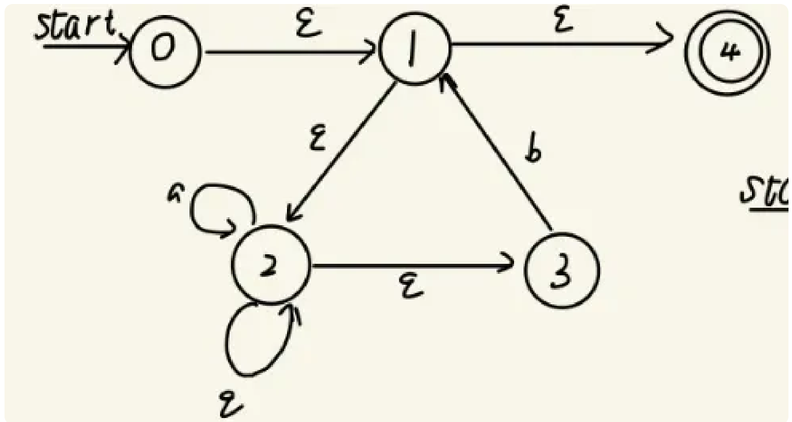


DFA

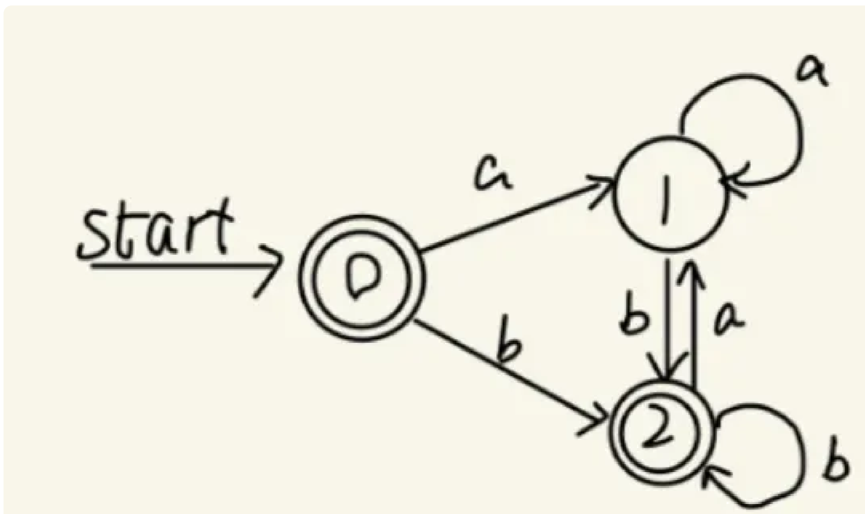
化简之后的唯一表示



2. L(((ϵ|a)*b)*)：空串，或者由字符{a,b}构成且以b结尾的字符串。

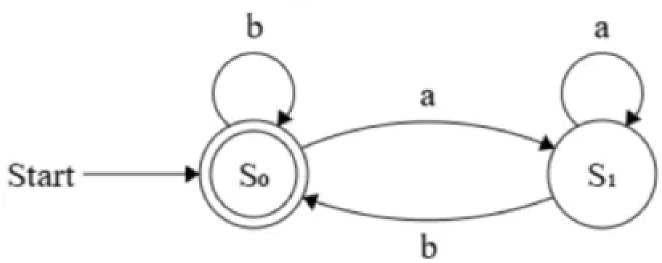


NFA

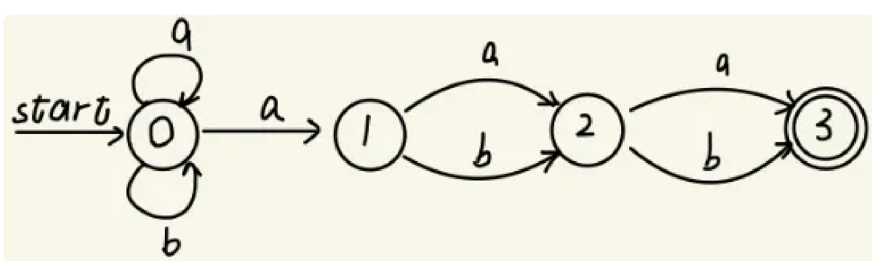


DFA

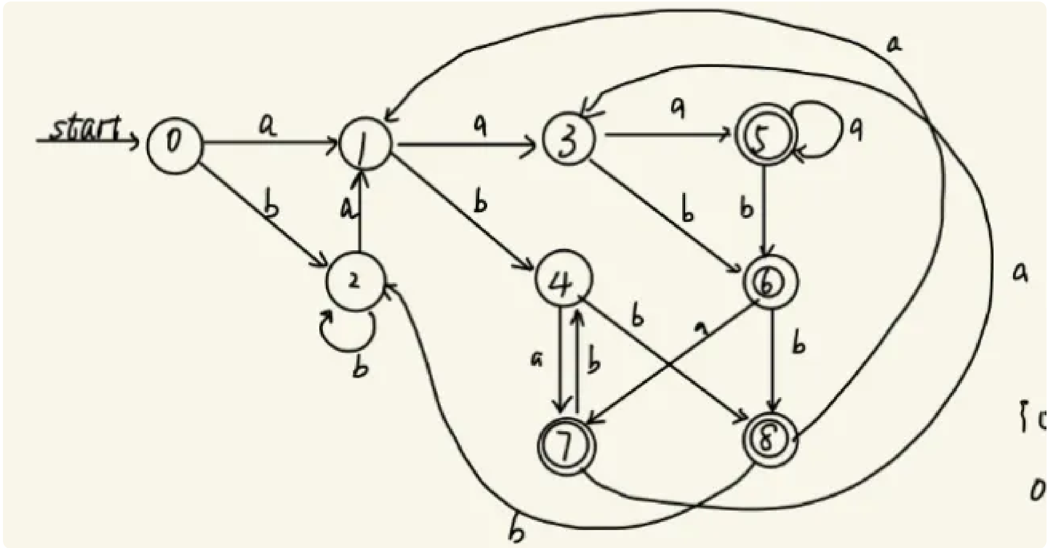
化简之后的唯一表示



3. L((a|b)*a(a|b)(a|b))：由字符{a,b}构成且以{aaa, aab, aba, abb}结尾。

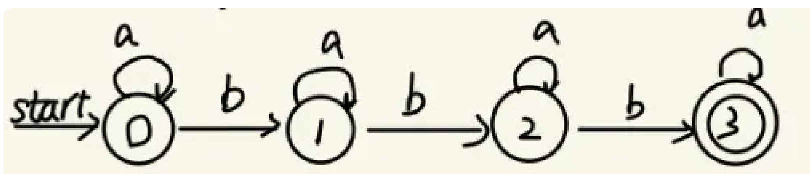


NFA



DFA

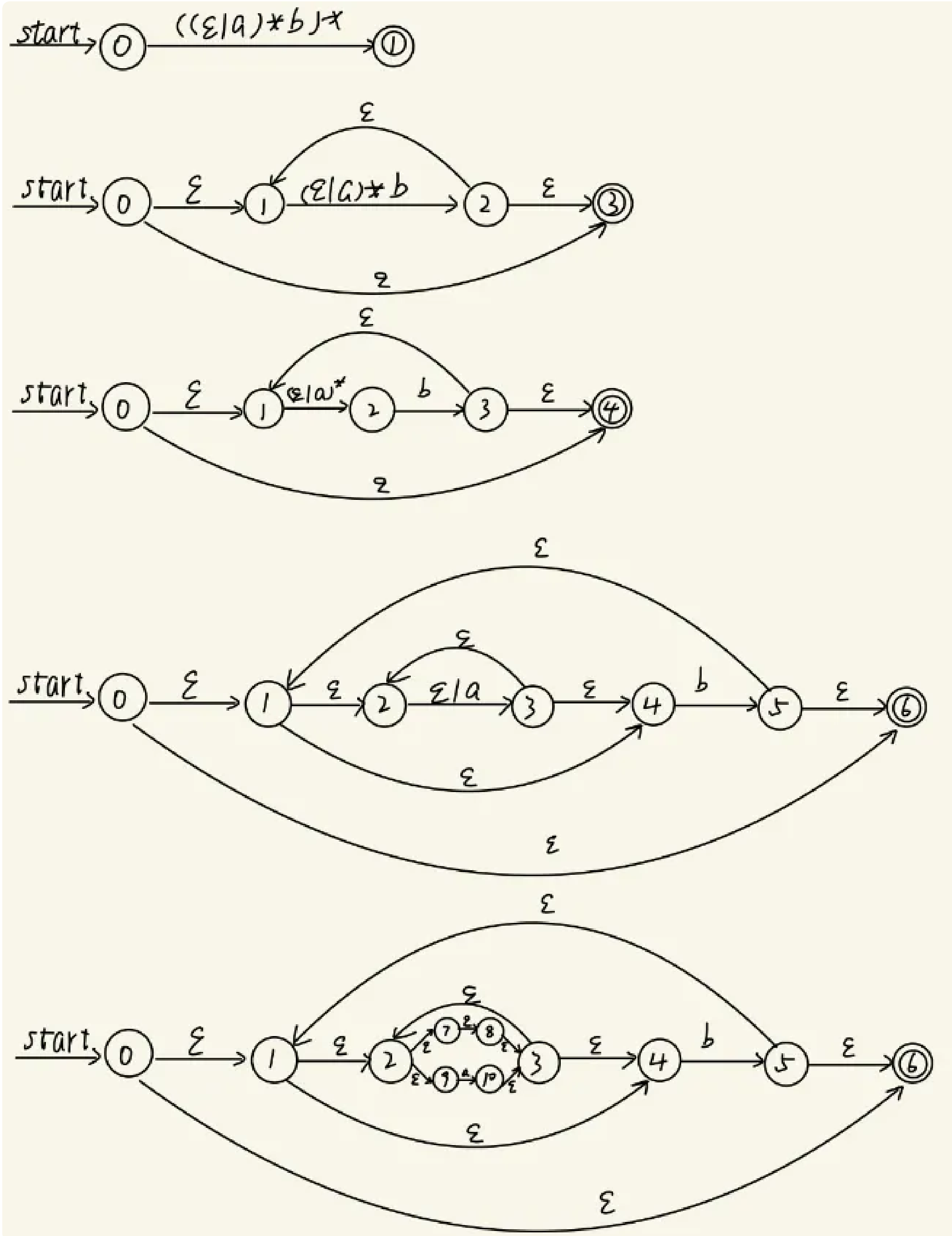
4. L(a*ba*ba*ba*)：由字符{a,b}构成且有且仅有3个b。



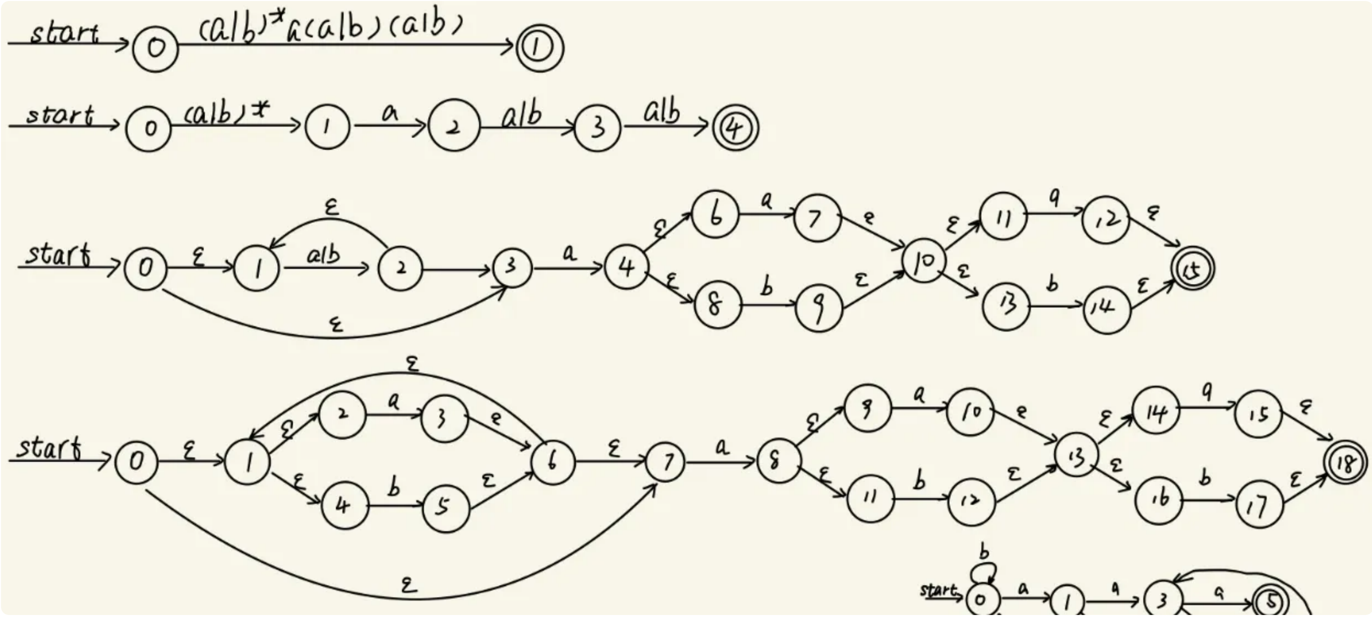
NFA and DFA

Exercise2

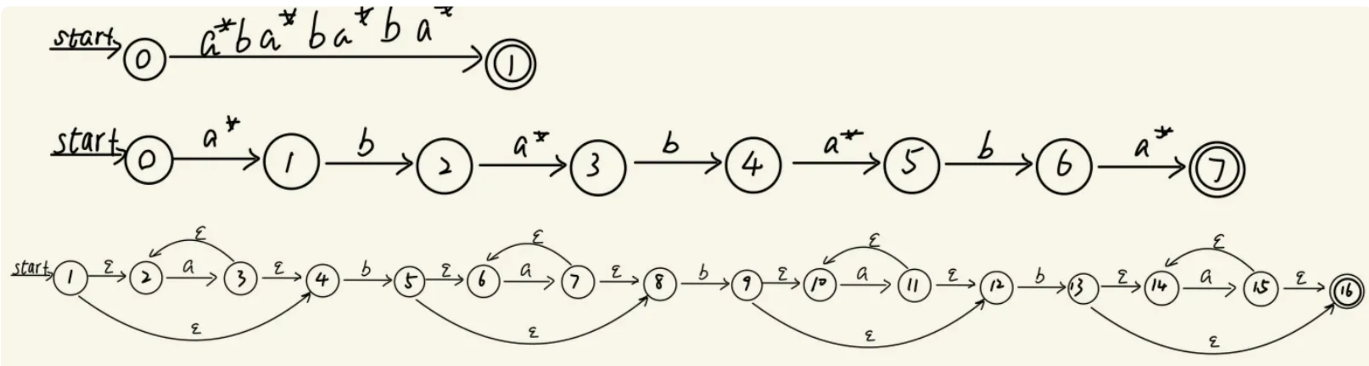
1. ((ϵ|a)*b)*


start
((ε|a)*b)*
start
ε
(ε|a)*b
ε
ε
ε
start
ε
ε|a*
b
ε
ε
ε
start
ε
ε
ε|a
ε
b
ε
ε
ε
ε
ε
start
ε
ε
ε
b
ε
ε
ε
ε
ε

2. (a|b)*a(a|b)(a|b)



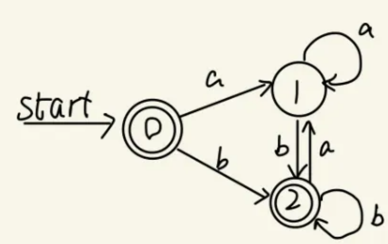
3. a*ba*ba*ba*



Exercise3

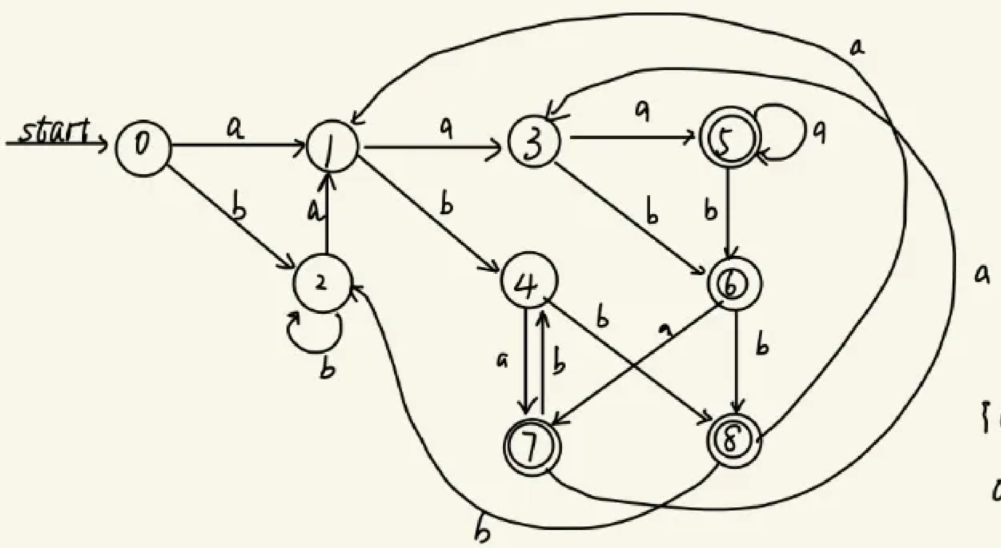
1. ((є|a)*b)*

I	a		b	
0 {0,1,2,7,8,3,4,6,9}	{10,3,4,2,7,8,9}	1	{5,6,1,2,7,8,3,9,4}	2
1 {10,3,4,2,7,8,9}	{10,3,4,2,7,8,9}	1	{5,6,1,2,7,8,3,9,4}	2
2 {5,6,1,2,7,8,3,9,4}	{10,3,4,2,7,8,9}	1	{5,6,1,2,7,8,3,9,4}	2



2. (a|b)*a(a|b)(a|b)

I	Ia	Ib
{0,1,2,4,7} 0	{1,2,3,4,6,7,8,9,11} 1	{1,2,4,5,6,7} 2
{1,2,3,4,6,7,8,9,11} 1	{1,2,3,4,6,7,8,9,10,11,13,14,16} 3	{1,2,4,5,6,7,12,13,14,16} 4
{1,2,4,5,6,7} 2	{1,2,3,4,6,7,8,9,11} 1	{1,2,4,5,6,7} 2
{1,2,3,4,6,7,8,9,10,11,13,14,16} 3	{1,2,3,4,6,7,8,9,10,11,13,~~14~~,15,16,18} 5	{1,2,4,5,6,7,12,13,~~14~~,16,17,18} 6
{1,2,4,5,6,7,12,13,14,16} 4	{1,2,3,4,6,7,8,9,~~11~~,15,18} 7	{1,2,4,5,~~6~~,7,17,18} 8
{1,2,3,4,6,7,8,9,10,11,13,~~14~~,15,16,18} 5	{1,2,3,4,6,7,8,9,10,11,13,14,~~15~~,16,18} 5	{1,2,4,5,6,7,12,13,~~14~~,16,17,18} 6
{1,2,4,5,6,7,12,13,~~14~~,16,17,18} 6	{1,2,3,4,6,7,8,9,11,~~15~~,18} 7	{1,2,4,5,~~6~~,7,17,18} 8
{1,2,3,4,6,7,8,9,~~11~~,15,18} 7	{1,2,3,4,6,7,8,9,10,11,13,14,16} 3	{1,2,4,5,6,7,12,13,14,16} 4
{1,2,4,5,6,17,~~1~~8} 8	{1,2,3,4,6,7,8,9,11} 1	{1,2,4,5,6,7} 2



{0,1,2,3,4}

a ↓ ↓ ↓ ↓ ↓

1 3 1 5 7

{0,1,2} {3,4}

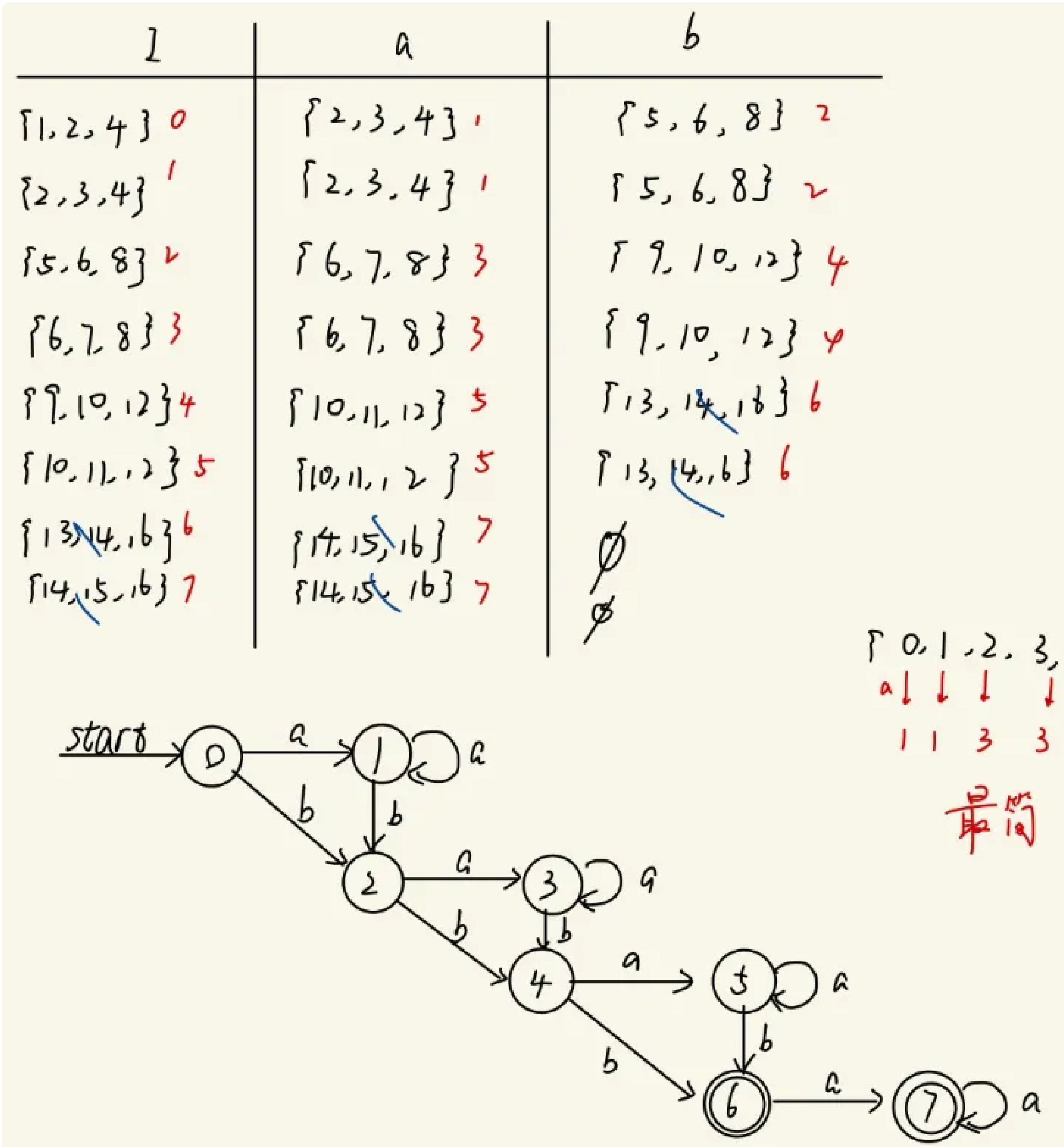
b ↓ ↓ ↓ ↓ ↓

2 4 2 6 8

{0,2} {1} {3} {4}

0 = 2

3. a*ba*ba*ba*

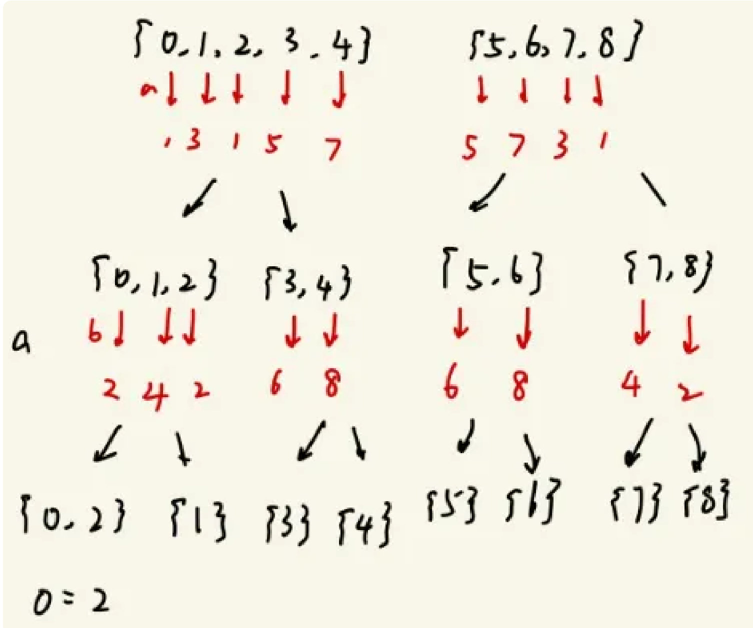


I	a	b
{1,2,4} 0	{2,3,4} 1	{5,6,8} 2
{2,3,4} 1	{2,3,4} 1	{5,6,8} 2
{5,6,8} 2	{6,7,8} 3	{9,10,12} 4
{6,7,8} 3	{6,7,8} 3	{9,10,12} 4
{9,10,12} 4	{10,11,12} 5	{13,14,16} 6
{10,11,12} 5	{10,11,12} 5	{13,14,16} 6
{13,14,16} 6	{14,15,16} 7	∅
{14,15,16} 7	{14,15,16} 7	∅

Bonus

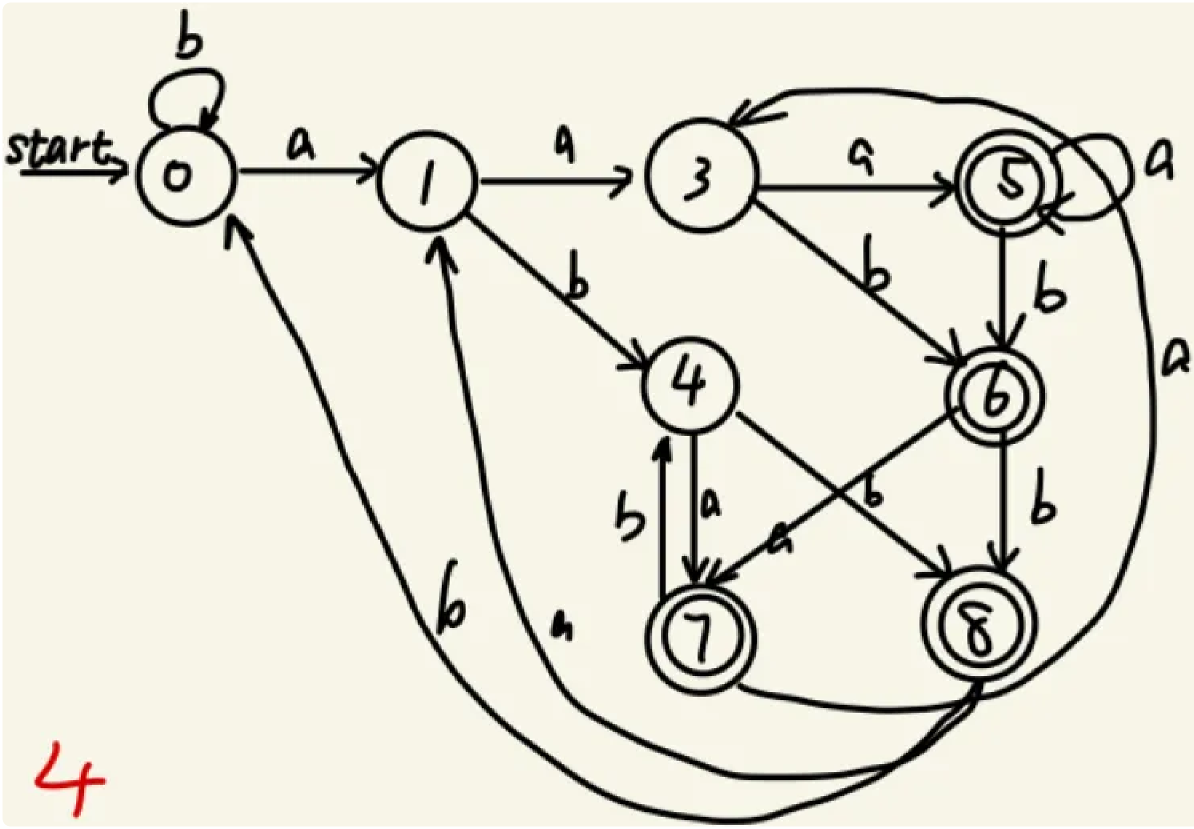
1. (a|b)*a(a|b)(a|b)

由算法可知，0和2是等价的。



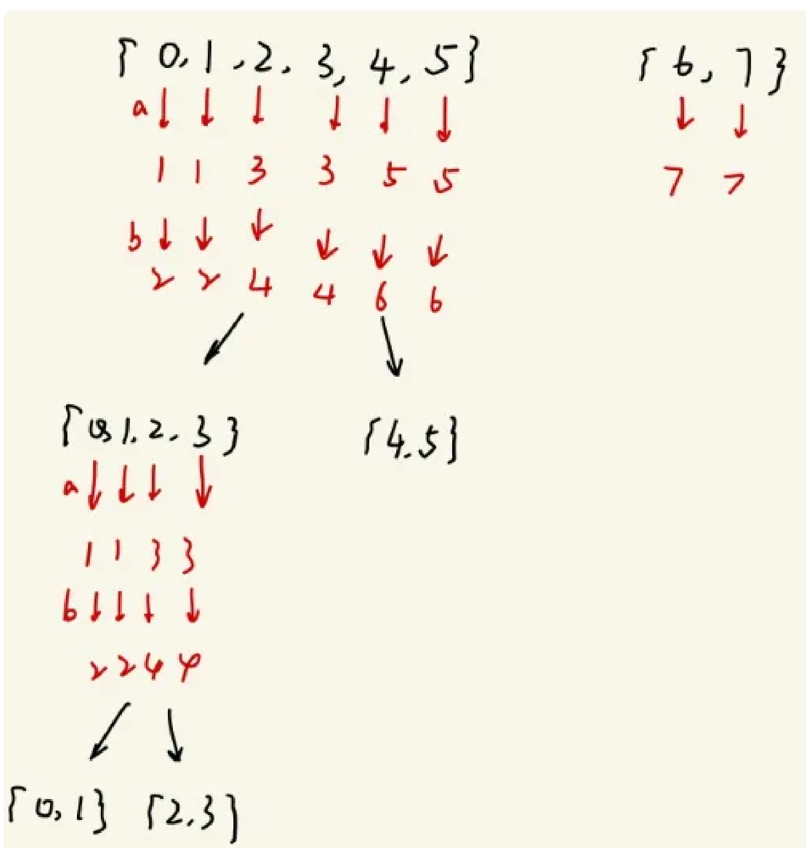
重新整理状态表得到。

I	Ia	Ib
0	1	0
1	3	4
3	5	6
4	7	8
5	5	6
6	7	8
7	3	4
8	1	0

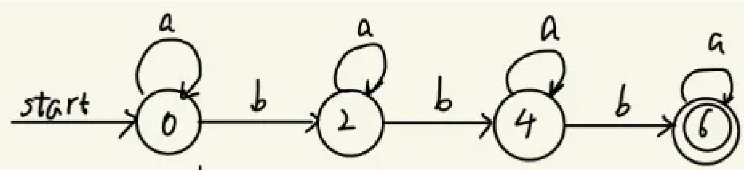


2. a*ba*ba*ba*

由算法可知，0和1,2和3，4和5以及6和7都是等价的。



重新整理状态表，得到化简后的DFA。



I	a	b
0	0	2
2	2	4
4	4	6
6	6	∅